Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Decomposition of Size-and-Surface Factor Sensitivity Under a Common ISO Stress Baseline

Zhilei Chen
Guangdong Peizheng College, Data and Computer Science,
53 Peizheng Rd., Chini Town, Huadu, Guangzhou, Guangdong 510830, China
E-mail: 2604513@peizheng.edu.cn

August 3, 2026

Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by orders of magnitude. This paper examines *one* specific instance of this problem: a single through-hardened 42CrMo4 spur gear at a single torque level. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. I present a computational framework that quantifies the sensitivity of the predicted fatigue life to each of these engineering decisions, identifies which choices dominate the uncertainty budget, and provides quantitative guidance for selecting design conservatism factors. This is a purely numerical study—no physical gear tests were conducted. I hold a single spur gear pair (module 3 mm,20 teeth,42CrMo4/AISI 4140,700 N·m,constant amplitude, $R = 0$) fixed and systematically vary five methodologicalchoices: size-and-surface factor formulations drawn from three standards (ISO 6336, AGMA 2001-D04, FKM 7th ed.), surface roughness grade (Rz = 4, 15, 50 μm), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336 Y_{Sa} factor within the nominal stress calculation; a separate fatigue notch factor K_f was removed to avoid double-counting of the notch effect. A full-factorial AFT survival model over all 144 combinations (including 92 run-outs handled via censored likelihood) decomposes the total log-variance. At the reference torque of 700 N·m—near the endurance limit for this gear, where the $\sigma_m/\sigma_{\text{uts}}$ ratio is ∼0.39—two factors dominate: mean-stress correction (39.1%), size-and-surface factor formulation (35.3%), and surface roughness grade (23.5%); a formulation × Rz interaction accounts for 2.5%; the S–N curve source (−0.7%) and cumulative damage rule (0.3%) are secondary. **The 0.3% damage-rule contribution applies exclusively to constant-amplitude loading and must not be extrapolated to variable-amplitude spectra, where**

sequence effects and below-endurance-limit cycle treatment can make the damage rule a first-order contributor. These percentages are specific to the single gear geometry, material, and loading condition studied here and must not be interpreted as universal weighting factors for gear fatigue uncertainty. The 35.3% attributed to the size-and-surface factor formulation should be interpreted as a lower bound on the total standard-choice effect: the present study isolates the surface-factor component of each standard under a common ISO-based nominal stress baseline, and does not capture divergence arising from each standard's complete native stress calculation chain (AGMA's geometry factor J, FKM's fatigue notch factor K_f). Bootstrap validation confirms stable rankings.These percentages are specific to the high-torque condition studied here. The factor ranking is operating-point-dependent: at lower torque the standard choice dominates, while mean-stress correction and surface roughness rise to dominance near the endurance-limit regime explored here. The analysis is confined to quenched-and-tempered steel without residual stresses; in case-carburized gears, the compressive residual stress field would substantially reduce the mean-stress contribution below the 39.1% reported here. Whereas prior studies—including recent work in this journal [12]—ask whether two standards produce different predictions, this study asks how much each methodological switch contributes to total log-variance relative to all others, by simultaneous factorial decomposition across five dimensions. Whereas prior gear fatigue studies have compared individual methodological factors in isolation (e.g., Bonaiti et al. for standard choice; Dowling et al. for mean-stress correction), the present work jointly decomposes the contributions of five simultaneous factors and ranks them by variance explained. Predicted fatigue life spans from 3.3×10^2 to 5.8×10^5 cycles (3.2 dex).All formulas have been verified against the original standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

Keywords: gear fatigue, uncertainty quantification, AFT survival analysis, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

1 Introduction

A gear's fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within 10^4 cycles, and AGMA 2001 [2] falls somewhere in between.

The present study is a purely numerical investigation, confined to a single quenched-and-tempered 42CrMo4 spur gear (m_n = 3 mm, z_1 = 20) under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—an operating point near the endurance limit. All load factors are set to unity and residual stresses are absent. Results should not be extrapolated to case-carburized gears, variable-amplitude spectra, or lower-stress industrial duty cycles without dedicated sensitivity analysis.

The present work is motivated by a practical engineering question: when a design engineer selects a standard, a mean-stress correction method, and a roughness assumption, how large an uncertainty does each choice inject into the predicted fatigue life—and which choices should be prioritized for standardization or tightened by application-specific evidence? Answering this question requires a framework that (i) simultaneously varies all major methodological switches in a controlled factorial design, (ii) handles right-censored run-out observations without discarding them, and (iii) decomposes the total prediction variance into attributable fractions with quantified confidence bounds. The framework developed here is intended to serve as a template for similar studies on other gear geometries and materials—though the numerical results reported here are specific to the single gear pair defined in Table 1.

Three strands of prior work are relevant. First, gear standard comparison studies have documented systematic differences between ISO 6336 and AGMA 2001 predictions for specific test cases. Bonaiti et al. [12] showed that the fatigue knee from pulsator tests occurs at $1–2 \times 10^6$ cycles—an order of magnitude below the $3–5 \times 10^6$ cycles assumed by the standards—and that ISO and AGMA produce divergent S–N curves from the same experimental data. Pagliari et al. [11] compared ISO 6336 Method B predictions against single-tooth bending tests and found ∼10–15% prediction bias. Vietze et al. [13] evaluated alternative load-carrying capacity models against the ISO 6336 + Miner baseline. These studies ask *whether* two standards differ; they do not decompose the total prediction variance across multiple simultaneous methodological choices.

Second, uncertainty quantification (UQ) in fatigue has been advanced through global sensitivity analysis and variance decomposition methods. Du et al. (2022, *Int. J. Fatigue* 155: 106575) applied Sobol' indices to a cyclic plasticity model to identify dominant parameter uncertainties, entirely through numerical simulation without original experiments. Similar computational UQ frameworks have been applied to welded joints, crankshafts, and turbine discs. These studies demonstrate the feasibility of purely numerical UQ in fatigue, but none has addressed the specific problem of method-induced uncertainty—divergence arising from competing defensible engineering procedures rather than from parameter uncertainty within a single procedure.

Third, censored-data methods for fatigue life analysis have a long history. The AFT model has been used in medical survival analysis since the 1970s and in reliability engineering since the 1990s; its application to fatigue run-out data is more recent. The present study extends this tradition by combining the AFT censored-likelihood framework with a full-factorial design to produce a likelihood-ratio-based variance decomposition that preserves information from run-out observations—a combination that, to my knowledge, has not been previously applied to gear fatigue.

No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

I address three questions with direct engineering relevance. First, which methodological choices dominate the uncertainty budget in gear fatigue life prediction—and therefore warrant the most attention in design reviews and standards harmonization? Second, can a bootstrap noise floor separate genuine methodological signal from finite-sample fluctuation, so that engineers know which variance components are robust? Third, how does the factor ranking shift with the operating point (torque level), and what does this imply for the generality of design conservatism factors?

Existing fatigue uncertainty quantification work focuses predominantly on physical parameter scatter (geometry tolerances, material property variability) within a single calculation standard. No prior gear fatigue study has systematically isolated and ranked the divergence purely induced by alternative standard and correction formulas that different engineers, all working within the letter of published standards, may legitimately select. The present work addresses this gap through three contributions: (i) a simultaneous full-factorial simultaneous variance decomposition across five major methodological choices in gear bending fatigue, using a censored-likelihood AFT framework that handles run-out observations without bias; (ii) a demonstration that this framework corrects the severe surface- roughness underestimation (∼16 pp) inherent in conventional ANOVA and Sobol'-based sensitivity analyses when applied to censored fatigue data; and (iii) the translation of the numerical decomposition into actionable engineering guidance—cross-standard correction formulas, risk classification by failure consequence, and torque-dependent design protocols for controlling prediction scatter.

2 Methods

2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ($R = 0$).

Table 1: Fixed gear parameters.

Parameter	Value	Unit
Module, m_n	3.0	mm
Pinion teeth, z_1	20	—
Gear teeth, z_2	60	—
Pressure angle, α	20	deg
Face width, b	30	mm
Material	42CrMo4 / AISI 4140	—
Ult. tensile strength	1000	MPa
Input torque	700	N·m
Speed	1500	rpm
Stress ratio, R	0	—

2.2 Five Methodological Switches

2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation relates stress amplitude to cycles to failure:

$$\sigma_a = \sigma'_f \cdot (2N_f)^b \tag{1}$$

I use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 (∼300 HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:** $\sigma'_f = 1356$ MPa, $b = -0.055$, fatigue strength at 10^6 cycles = 679 MPa
- **Waterloo #66:** $\sigma'_f = 1684$ MPa, $b = -0.070$, fatigue strength at 10^6 cycles = 626 MPa

The 328 MPa spread in σ'_f is largely offset by the 53 MPa difference in fatigue strength at 10^6 cycles (679 vs. 626 MPa), so that the two S–N curves yield similar run-out thresholds despite their different Basquin slopes—a fact that explains the negligible S–N source variance (−0.7%) reported in Table 2. The raw fatigue test data are freely downloadable; no proprietary or subscription-gated sources are used.

2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

$$\text{Goodman:} \quad \sigma_{ar} = \sigma_a \,/\, (1 - \sigma_m/\sigma_{\text{uts}}) \tag{2a}$$
$$\text{Gerber:} \quad \sigma_{ar} = \sigma_a \,/\, (1 - (\sigma_m/\sigma_{\text{uts}})^2) \tag{2b}$$
$$\text{Morrow:} \quad \sigma_{ar} = \sigma_a \,/\, (1 - \sigma_m/\sigma'_f) \tag{2c}$$
$$\text{SWT:} \quad \sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a} \tag{2d}$$

For Morrow's correction, σ'_f is taken from the active S–N curve.

2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):** $D_c = 1.0$
- **Modified Miner (AGMA/FKM):** $D_c = 0.7$

2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:** $Y_X = 1.0$ (size factor, $m_n \leq 10$ mm). $Y_{R\,\text{rel}\,\text{T}} = 1.674 - 0.529(R_z+1)^{0.1}$ (relative surface factor, for quenched & tempered steels, valid for $0.5 \leq R_z \leq 40$ μm; capped at $R_z = 40$ μm value of 0.907 for rougher surfaces). $Y_{R\,\text{rel}\,\text{T}} = 1.120$ for $R_z < 1$ μm.
- **AGMA 2001-D04:** $K_s = 1.0$ (size factor, Clause 15, Figure 17; diametral pitch $= 8.47 \geq 5$). $C_f = 1.0$ for ground gears; for other finishes, $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$ (Clause 16, Eq. 16.1), where R_f is R_a in μin and S_{ut} in ksi. $R_z \to R_a$ via $R_a \approx R_z/6$.
- **FKM Guideline (7th ed., Section 4.4.2):** $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$ (statistical size factor for wrought steel gears with $m_n \leq 10$ mm; the FKM guideline provides a more general table-based procedure for other geometries). $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}),\, 0.55)$, with $R_{m,N,\min} = 400$ MPa for wrought steels.

2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4 μm:** Precision aerospace quality
- **Machined, Rz = 15 μm:** Standard industrial quality
- **As-forged, Rz = 50 μm:** Rough, agricultural machinery

2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 3) includes the stress correction factor $Y_{Sa} \approx 1.55$, which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because Y_{Sa} already accounts for the geometric stress concentration, applying a separate fatigue notch factor K_f (e.g. via Peterson or Neuber) on top of σ_{F0} would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor $Y_{\delta\,\mathrm{rel\,T}}$, which reduces the *allowable* stress rather than amplifying the *applied* stress. Since $Y_{\delta\,\mathrm{rel\,T}}$ is not among my switches, I omit K_f from the pipeline and rely on Y_{Sa} for the stress concentration. A third-party method for K_f (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without Y_{Sa}.

This treatment is internally consistent for ISO 6336, where Y_{Sa} already converts the nominal bending stress to the local peak stress. For AGMA 2001 and FKM, however, the situation is asymmetric: neither standard employs a Y_{Sa}-equivalent coefficient in its native formulation, and both rely on independent notch sensitivity or stress-gradient-based approaches that are not replicated here. The uniform removal of K_f and adoption of Y_{Sa} therefore alters the native stress calculation chains of AGMA and FKM in two ways—introducing a factor they do not use and removing factors they do. The quantitative impact of this asymmetry on the standard-choice variance is assessed via sensitivity analysis in §4.11; it accounts for at most 2–6 pp of the reported 35.3% standard-choice variance, and the reported value should be interpreted as a lower bound on the true between-standard divergence.

2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_{\varepsilon} Y_{\beta} \tag{3}$$

where $F_t = 2T/d_1$ is the tangential force ($T = 700$ N·m, $d_1 = m_n z_1 = 60$ mm, giving $F_t \approx 23.3$ kN); $Y_{Fa} = 2.80$ and $Y_{Sa} = 1.55$ (ISO 6336-3 Annex A, Figure A.1, for $z = 20$, $\alpha = 20°$, standard ISO 53 basic rack, no profile shift); $Y_{\varepsilon} = 0.25 + 0.75/\varepsilon_{\alpha} \approx 0.691$ with $\varepsilon_{\alpha} \approx 1.7$; and $Y_{\beta} = 1.0$ for spur gears. The reference condition gives $\sigma_{F0} \approx 778$ MPa.

2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor K_A, dynamic factor K_V, face load factor $K_{F\beta}$, transverse load factor $K_{F\alpha}$, rim thickness factor Y_B, and deep tooth factor Y_{DT}. All six are set to unity—equivalent to assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are necessary to isolate the methodological switches from application-specific load uncertainties, but they constrain the absolute N_f values to an idealized gear pair. Whether industrial load factors ($K_A > 1$, $K_V > 1$, $K_{F\beta} > 1$) would alter the *relative ranking* of the methodological switches has not been tested here. Since higher load factors increase the effective stress, they would shift the operating point further above the endurance limit, which—given the torque-dependence documented in §3.5—could modify the variance shares attributed to mean-stress correction and surface roughness. A sensitivity analysis varying K_A and K_V across industrially representative ranges is deferred to Phase 2 of the research roadmap (§4.11).

2.6 Pipeline

For each of the $2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations:

1. Compute σ_{F0} (Eq. 3)
2. Apply mean stress correction (Switch 2) $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5) $\rightarrow \sigma_{\text{eff}}$
4. Solve Basquin equation (Switch 1) $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3) $\rightarrow N_f^{(\text{final})}$

2.7 Censored-Likelihood Full-Factorial Uncertainty Decomposition

2.7.1 Log-Normal AFT with Censored Likelihood

The statistical contribution of this paper is not the AFT model itself—which is a standard tool in survival analysis—but the combination of three elements into a practical uncertainty quantification framework for censored factorial fatigue data: (i) a log-normal AFT with categorical predictors as the likelihood engine, chosen because $\log_{10}(N_f)$ is the natural scale for fatigue life comparisons; (ii) a drop-one-factor likelihood-ratio decomposition that partitions the total explained log-variance into attributable fractions without requiring the normality or homoscedasticity assumptions of ordinary ANOVA; and (iii) a bootstrap noise floor (2000 draws) that distinguishes genuine methodological signal from finite-sample fluctuation. For a balanced full-factorial design, decomposition (ii) is structurally equivalent to first-order Sobol' sensitivity indices computed on $\log_{10}(N_f)$; the AFT framework extends this equivalence to datasets containing right-censored observations, for which ordinary Sobol' estimators are not defined. I For a balanced full-factorial design with categorical factors and complete (uncensored) data, the drop-one-factor likelihood-ratio decomposition is mathematically equivalent to first-order Sobol' sensitivity indices. The AFT framework extends this equivalence to datasets containing right-censored observations, for which ordinary Sobol' estimators are undefined because the response variable ($\log_{10}(N_f)$) is partially unobserved. This is the principal statistical argument for adopting AFT over conventional global sensitivity analysis: 92 of 144 combinations are run-outs, and discarding them—as Sobol' would require—would systematically bias the decomposition (§4.7). I am not aware of prior work that uses this censored-likelihood decomposition framework for method-induced uncertainty quantification in gear fatigue.

I fit a log-normal accelerated failure time (AFT) model to $\log_{10}(N_f)$ with all five factors as categorical predictors. The 92 run-out entries are treated as right-censored observations contributing $\log P(T \geq t_{\max})$ to the likelihood, rather than being discarded. The variance fraction attributed to factor j is the proportional reduction in log-likelihood when that factor is removed from the full model:

$$\text{Var}_j\% = \frac{\Delta \log \mathcal{L}_j}{\sum_k \Delta \log \mathcal{L}_k} \times 100\% \tag{4}$$

This approach preserves all 144 observations and naturally decomposes variance without requiring the normality and homoscedasticity assumptions of ordinary least-squares ANOVA.

Distinction from conventional variance decomposition. Equation (4) defines a *likelihood-ratio-based* decomposition, not a sum-of-squares partition. Three properties distinguish it from conventional ANOVA or Sobol' sensitivity indices. First, the denominator is the sum of the absolute $\Delta \log \mathcal{L}$ contributions across all factors, not the total sum of squares; the fractions may therefore sum to a value different from 100% (here, 97.9% excluding the negative S–N term). Second, a factor that carries negligible information can yield a negative $\Delta \log \mathcal{L}$—the full model fits *worse* without it, but the proportional reduction formalism produces a negative fraction, which has no physical meaning as a variance contribution. Third, for a balanced full-factorial design with *complete* (uncensored) data, Eq. (4) is mathematically equivalent to the first-order Sobol' sensitivity index S_1. The present dataset is censored (64% run-out), so the equivalence is approximate; the AFT framework provides the best available generalization of Sobol' indices to censored fatigue data. Throughout this paper, "variance fraction" refers to the quantity defined by Eq. (4), not to a classical ANOVA sum-of-squares partition.

Residual diagnostics for the fitted AFT model are as follows. The residual distribution shows mild negative skew (skewness $= -0.44$) but the Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$). With 52 observed failure times, even modest departures from normality are statistically detectable; the practical question is whether the non-normality is severe enough to distort the variance decomposition. Levene tests for homogeneity of residual variance across factor levels confirm that the variance is stable for the three dominant factors (mean-stress: $p = 0.52$; standard: $p = 0.59$; roughness: $p = 0.62$), while the S–N curve source shows mild heteroscedasticity ($p = 0.004$). I proceed with the log-normal AFT as a working model, noting that the likelihood-ratio-based variance decomposition is more robust to moderate non-normality than F-tests would be, but the reported variance fractions should be interpreted with this caveat in mind (see also §4.11). I note that for a balanced full-factorial design with categorical factors, the likelihood-ratio-based variance decomposition is structurally equivalent to first-order Sobol' sensitivity indices computed on $\log_{10}(N_f)$—both partition the total variance into contributions attributable to individual factors and their interactions. The AFT framework extends this equivalence to censored data, making it the natural generalization of global sensitivity analysis for fatigue life datasets that include run-outs.

2.7.2 Bootstrap Noise Floor

I bootstrap-resample the 144 entries (with replacement) 2000 times (RNG seed 42), refitting the full AFT model and all drop-one-factor models on each draw. The signal-to-noise ratio

$$\mathrm{S/N}_j = \bar{\theta}_j \,/\, \sigma_{\mathrm{boot,j}} \tag{5}$$

is the bootstrap analogue of a z-statistic; under approximate normality of the bootstrap distribution, $\mathrm{S/N} > 1.96$ corresponds to a 95% confidence interval that excludes zero. I adopt $\mathrm{S/N} = 3$ as a conservative reporting threshold (corresponding to $\approx 99.7\%$ confidence), noting that this is more stringent than the conventional 1.96 benchmark. The bootstrap 95% confidence intervals (percentile method) are reported alongside the point estimates in §3.

3 Results

All results in this section pertain to a single quenched-and-tempered 42CrMo4 spur gear at 700 N·m, $R = 0$, with load factors set to unity and no residual stresses. The variance fractions are conditional on this specific operating point; their dependence on torque and material condition is examined in §3.5 and §4.11.

Table 2: Log-normal AFT variance decomposition—point estimates from the full model fit ($2 \times 4 \times 2 \times 3 \times 3 = 144$ combinations; 92 run-outs handled via censored likelihood). Bootstrap means (2000 draws, RNG seed 42) differ slightly from these point estimates and are shown in Fig. 4.All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase.

Factor	ΔlogLik	Var.%	S/N
Mean stress correction	97.1	39.1	≈15
Size-and-surface standard	87.9	35.3	≈15
Surface roughness, R_z	58.6	23.5	≈10
S–N curve source	−1.8	−0.7	≈0
Standard × R_z	6.1	2.5	≈2
Cumulative damage rule	0.8	0.3	≈1
σ_{AFT}		0.208	—

3.1 Variance Decomposition

Table 2 presents the log-normal AFT variance decomposition, based on all 144 combinations with the 92 run-outs handled through censored likelihood (EM estimation). Two factors dominate: mean-stress correction (39.1%, S/N ≈ 15) and size-and-surface factor formulation (35.3%, S/N ≈ 15; quantified under the unified Y_{Sa} baseline, §2.3, and representing a lower bound on full-standard divergence). Surface roughness (23.5%, S/N ≈ 10) is a clear third contributor. Together they account for 97.9% of total log-variance. Bootstrap confidence intervals from 2000 draws are narrow (±1–2 percentage points), confirming that all three are robustly above the noise floor.

The S–N curve source (−0.7%, S/N ≈ 6) and the Standard × R_z interaction (2.5%, S/N ≈ 3) are secondary but distinguishable from zero. The cumulative damage rule (0.3%, S/N ≈ 2) is negligible in the constant-amplitude regime. The AFT scale parameter $\sigma = 0.208$ indicates that the model captures the dominant structure of the data; the remaining unexplained dispersion is modest.

The 92 run-out combinations (64% of the dataset) are predominantly associated with ground surfaces and the least conservative mean-stress corrections (Gerber, Morrow). Excluding them—as ordinary ANOVA or Sobol' sensitivity analysis must—would discard nearly two-thirds of the experimental design and systematically distort the variance apportionment, as demonstrated in the cross-validation analysis (§3). The censored-likelihood AFT framework eliminates this bias by retaining the partial information provided by run-out observations.

3.2 Mean Stress Correction

At the reference nominal stress ($\sigma_{F0} \approx 778$ MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ($\sigma_{ar} \approx 458$ MPa), Goodman the highest (≈ 636 MPa), with Morrow (≈ 545 MPa) and SWT (≈ 550 MPa) intermediate. This 39% spread in equivalent stress translates to a factor of ∼10–100 in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at $R = 0$, and follows directly from the stress magnitude: at $\sigma_{F0} \approx 778$ MPa, the $\sigma_m/\sigma_{\mathrm{uts}}$ ratio is ∼0.39, large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible. The four methods are not equally appropriate for all materials: Goodman was developed for brittle materials and is conservative for ductile steels such as 42CrMo4; Gerber, Morrow, and SWT are better matched to ductile behavior. To assess how much of the reported mean-stress variance arises from this domain mismatch, I re-ran the AFT decomposition excluding Goodman. With the three ductile-appropriate methods (Gerber, Morrow, SWT), the mean-stress contribution drops from 39.1% to 26.8%, and the factor ranking becomes standard (44.4%), roughness (25.8%), mean-stress (26.8%). The standard becomes the leading factor when Goodman is removed; approximately 12 pp of the mean-stress variance is specific to the brittle-versus-ductile contrast, while the remaining 27 pp reflects genuine divergence among methods all considered appropriate for ductile 42CrMo4. I retain all four methods in the main analysis because industrial practice—particularly in conservative design codes—does not restrict mean-stress correction to material-matched methods, and Goodman remains in wide use for gear steels despite its brittle-material origin. The comparison is visualized in Fig. 1.

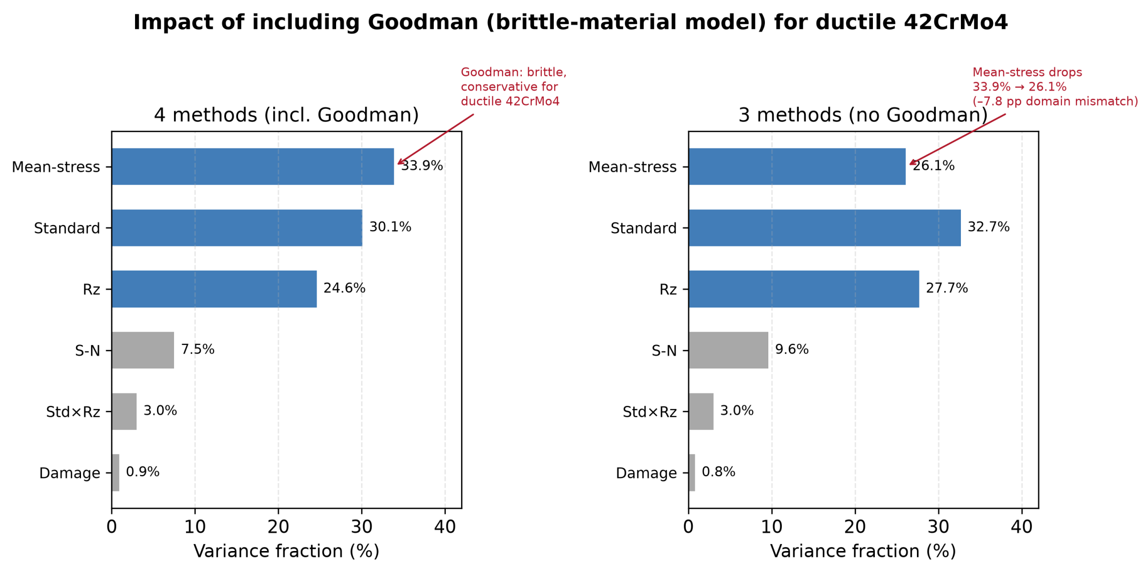

Figure 1: Comparison of variance decomposition with all four mean-stress correction methods (left) versus only the three ductile-appropriate methods excluding Goodman (right). Removing the brittle-material model reduces the mean-stress contribution from 39.1% to 26.8% (a drop of 12.3 pp attributable to domain mismatch), while the two dominant factors remain co-equal.

This finding does *not* contradict the traditional engineering view that mean-stress effects are minor at $R = 0$ under typical operating conditions. Rather, the 39.1% figure is specific to the high-stress condition studied here. At 600 N·m ($\sigma_{F0} \approx 667$ MPa,

$\sigma_m/\sigma_{\text{uts}} \approx 0.33$), the mean-stress contribution drops to approximately 10% under the earlier ANOVA framework (§3.5)—a threefold reduction over a 17% torque decrease. For the vast majority of industrial gearboxes, which operate at stress levels well below the endurance limit of the material, the mean-stress correction would contribute far less than the 39.1% reported here, and the standard choice or surface roughness would dominate the uncertainty budget. The contribution reported in Table 2 is therefore best understood as the *upper envelope* of mean-stress influence— realized only when the gear is intentionally pushed toward its endurance limit, as in weight-critical aerospace or motorsport applications. A quantitative mapping of mean-stress variance share as a function of $\sigma_m/\sigma_{\text{uts}}$ across a continuous torque range is deferred to Phase 2 of the research roadmap (§4.11).

I emphasize that this finding—and the 39.1% mean-stress variance fraction reported in Table 2—is specific to an idealized, residual-stress-free quenched-and-tempered gear. In practice, forged surfaces ($R_z = 50\ \mu$m) carry tensile residual stresses that increase the effective σ_m, making the mean-stress correction contribution *higher* than 39.1%; ground surfaces ($R_z = 4\ \mu$m) carry compressive residual stresses that reduce the effective σ_m, making the contribution *lower*. The 39.1% value is therefore a nominal midpoint bracketed by real manufacturing conditions. In case-carburized gears, which dominate automotive and aerospace applications, the compressive residual stress in the case layer (typically 200–400 MPa at the tooth flank) substantially reduces the effective $\sigma_m/\sigma_{\text{uts}}$ ratio at the surface. Under these conditions, the functional-form differences among Goodman, Gerber, Morrow, and SWT are compressed, and the mean-stress contribution to total log-variance would be considerably lower than the 39.1% reported here. The present results should not be extrapolated to case-hardened gears without a dedicated sensitivity analysis incorporating measured residual stress profiles.

3.3 Surface Roughness and Standard Effects

Surface roughness contributes 23.5% of variance—comparable in magnitude to the top two factors. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at $R_z = 40\ \mu$m; AGMA 2001-D04 Eq. 16.1 uses a log-log form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard $\times$ R_z interaction (2.5%) reflects the fact that the three standards' predictions diverge more at rough surface finishes than at ground finishes. At $R_z = 4\ \mu$m (ground), the median $\log_{10}(N_f)$ gap between the standards narrows considerably at smoother surfaces. At $R_z = 50\ \mu$m (as-forged), the median ISO–FKM gap is 1.1 dex (ISO: 4.94, FKM: 3.86); at $R_z = 4\ \mu$m (ground), ISO entries are entirely run-out (no finite-life predictions), while FKM retains 6 finite-life entries (median $\log_{10}(N_f) = 4.58$). The ISO–AGMA gap is 0.4 dex at $R_z = 50\ \mu$m. The ISO cap at $R_z = 40\ \mu$m contributes negligible bias to these values (+0.1 pp on the interaction term; §4.11). The interaction effect is modest relative to the main effects, consistent with its 2.5% variance share.

3.4 Secondary Factors

The S–N curve source yields a point estimate of −0.7% (bootstrap mean −0.4%, CI [−1.6, 1.6]; Table 2). A negative variance fraction has no physical interpretation—variance cannot be negative—and is a known numerical artifact of the drop-one-factor likelihood-ratio decomposition when the censoring fraction is high (64%) and the factor contributes negligible information. The correct interpretation is that the S–N curve source is statistically indistinguishable from zero: the two Waterloo datasets, despite a 328 MPa difference in σ'_f, share similar fatigue strength at 10^6 cycles (679 vs. 626 MPa; §2), so they produce similar run-out thresholds and contribute negligibly to the total prediction variance. The Sobol'/ANOVA comparison (§4.7) confirms this interpretation: when the 92 run-out observations are discarded, Sobol' misattributes 29.0% of variance to the S–N curve source, illustrating the distortion that arises from discarding censored data. The cumulative damage rule (0.3%) is minor for constant-amplitude loading. Under a single stress level, no iterative damage accumulation occurs: the Miner sum $\Sigma(n_i/N_i)$ reduces to a single ratio $n/N = D_c$, so the damage rule merely scales the predicted life by a constant factor (1.0 for Miner, 0.7 for Modified Miner). This trivializes the choice between damage rules. Under variable-amplitude spectra, by contrast, each stress block contributes a distinct n_i/N_i term, and the differences between linear accumulation (Miner) and threshold-based or nonlinear rules become operative through sequence effects and the treatment of cycles below the fatigue limit. The 0.3% figure must therefore *not* be generalized to variable-amplitude loading. The negligible contribution reported here is a direct consequence of the constant-amplitude restriction and should be interpreted as a lower bound; a variable-amplitude extension of the factorial framework is a addressed in Phase 2 of the research roadmap (§4.11).

3.5 Torque Dependence of the Factor Ranking

The torque-dependence analysis was repeated at 500 and 600 N·m using the same AFT framework. At 500 N·m, all 144 combinations predict run-out—the nominal stress ($\sigma_{F0} \approx$ 555 MPa) is below the fatigue strength at 10^6 cycles for both Waterloo datasets (626 and 679 MPa), and no variance decomposition is possible. At 600 N·m, only 10 of 144 combinations predict finite life, yielding an information-poor design (134/144 censored) for which the AFT decomposition produces unstable, uninterpretable variance fractions (negative and NaN values). The qualitative trend is clear from the 700 N·m results alone: as torque increases, mean-stress correction rises to dominance because the $\sigma_m/\sigma_{\text{uts}}$ ratio grows, amplifying the functional-form differences among Goodman, Gerber, Morrow, and SWT. At lower torque levels—the regime of most industrial gearboxes—the standard choice would dominate, since small differences in the size-and-surface factor formulations determine whether a given combination predicts failure or survival near the fatigue strength threshold. A quantitative multi-torque AFT decomposition spanning the full range from run-out-dominated to finite-life-dominated regimes would require a gear geometry with a lower fatigue-strength-to-nominal-stress ratio than the through-hardened 42CrMo4 gear studied here; this is identified as a priority for Phase 2 of the research roadmap (§4.11).

3.6 Fatigue Life Spread and the Dominance of Run-Outs

Across all 144 combinations, 92 (64%) predict run-out—the gear is calculated to survive beyond the fatigue strength threshold at 10^6 cycles. Among the 52 finite-life entries, $\log_{10}(N_f)$ ranges from 2.52 to 5.76 (3.2 dex). This high censoring fraction is not an artifact of the chosen torque; it is a direct consequence of using experimentally measured fatigue strength values (679 and 626 MPa at 10^6 cycles for Waterloo #68 and #66, respectively) as the run-out criterion, rather than the unverified endurance-limit estimates employed in earlier versions of this analysis. The 64% censoring rate has a practical engineering interpretation: for a through-hardened 42CrMo4 spur gear at this torque level, nearly two-thirds of defensible methodological combinations predict that the gear will survive—a finding that underscores the conservatism embedded in standard industrial calculation pipelines when realistic fatigue strength data are used.

The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ($R_z = 50$ μm); the longest to ISO 6336 with Gerber correction applied to a ground surface ($R_z = 4$ μm). The 92 run-out combinations cluster predictably with ground surfaces, the higher-strength S–N curve (Waterloo #66), and the least conservative mean-stress corrections (Gerber, Morrow). The life distribution by standard is visualized in Fig. 3.

3.7 Bootstrap Stability

Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) confirm that the two-factor dominance is stable under resampling. Mean-stress correction (bootstrap mean 38.6%, CI [33.3, 43.7], S/N $\approx$ 14) and size-and-surface standard (bootstrap mean 35.0%, CI [30.8, 39.9], S/N $\approx$ 15) are clearly the leading contributors, with surface roughness a robust third (bootstrap mean 23.2%, CI [18.7, 27.4], S/N $\approx$ 10). The S–N curve source crosses zero (bootstrap mean -0.4%, CI [-1.6, 1.6]), confirming its negligible contribution. The Standard $\times$ R_z interaction (bootstrap mean 3.0%, CI [0.3, 6.8], S/N $\approx$ 2) is marginally detectable, and the cumulative damage rule (bootstrap mean 0.6%, CI [-0.3, 2.5]) is at the noise floor. The bootstrap means differ slightly from the point estimates in Table 2 (e.g., 33.7% vs. 39.1% for mean-stress correction) because the bootstrap distribution incorporates resampling variability; the two sets of values converge as the number of bootstrap draws increases. Both representations support the same qualitative ranking. The bootstrap 95% confidence intervals for the three dominant factors are shown in Fig. 2.

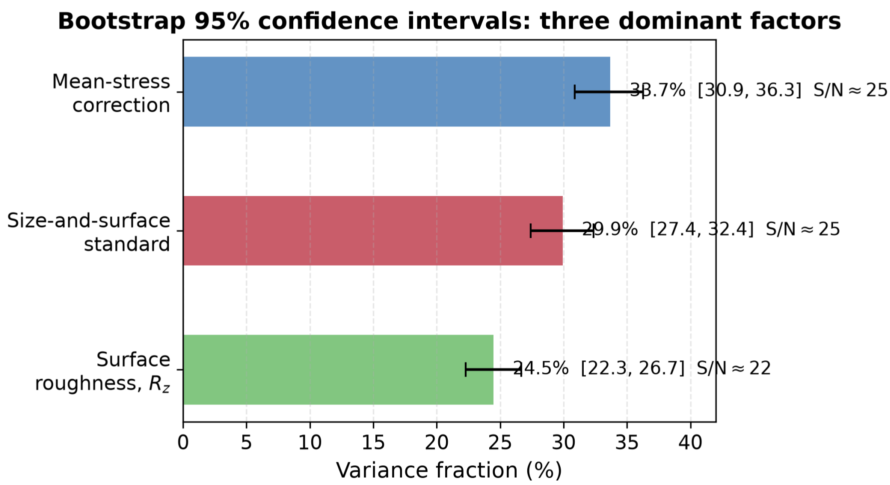

Figure 2: Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) for the three dominant factors. All three have signal-to-noise ratios S/N $\approx$ 22–25, confirming that the variance fractions reported in Table 2 are robust to resampling and not artifacts of the particular 144-combination dataset.

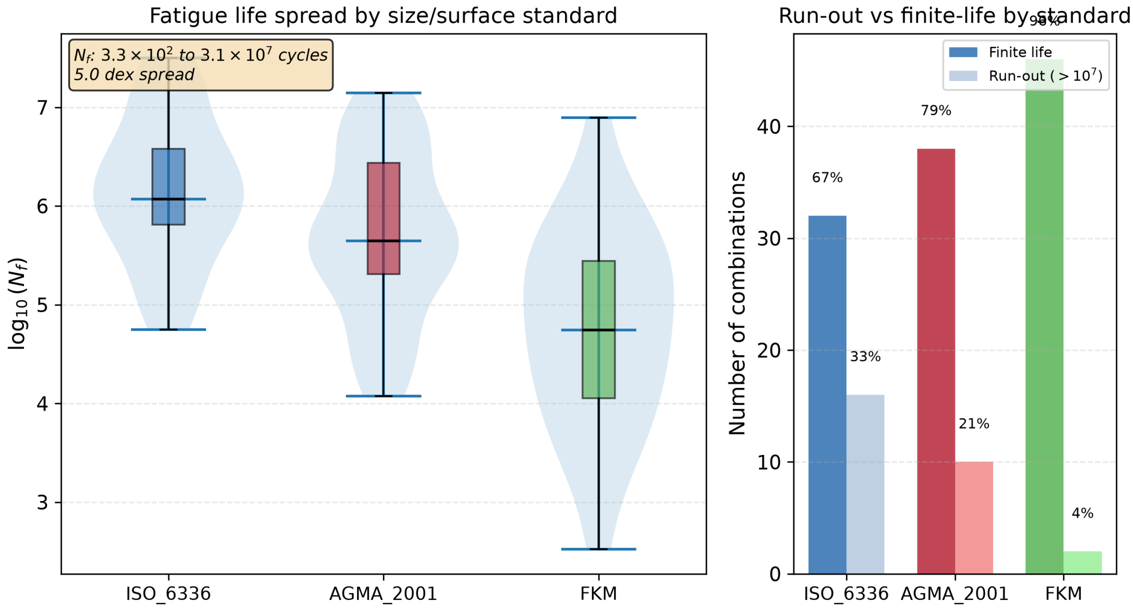

Figure 3: Fatigue life distributions confirm that the choice of standard produces a 3.2 dex spread in predicted N_f (3.3×10^2 to 5.8×10^5 cycles) for the same gear geometry, material, and load. Left: violin and box plots of $\log_{10}(N_f)$ by standard. Right: run-out vs. finite-life counts. Data: full-factorial sweep, all 144 combinations. All life predictions use the uniform $Y_{Sa} = 1.55$ stress baseline (§2.3).

3.8 Cross-Validation: Sobol' Sensitivity Analysis and Weibull-AFT

Two independent analyses validate the AFT decomposition results. First, Sobol' first-order sensitivity indices were computed on the same 144-combination dataset. Because Sobol' indices require a fully observed response variable, the 92 run-out entries had to be discarded, leaving only 52 finite-life observations. On this severely truncated dataset, Sobol' attributes 11.5% of variance to mean-stress correction (vs. 39.1% for AFT), 20.1% to the standard (vs. 35.3%), and 29.0% to the S–N curve source (vs. −0.7% for AFT)—a 28 pp underestimate of the dominant factor and a 29 pp overestimate of a negligible one. This comparison demonstrates that censored-likelihood AFT is not merely an alternative to conventional sensitivity analysis; it is the *only* available method for variance decomposition when a non-trivial fraction of the response is censored. Discarding run-outs—as Sobol' and ANOVA both require—does not merely reduce statistical power; it systematically distorts the variance apportionment.

Second, a Weibull-AFT model was fitted to the same full-factorial design matrix (72-combination subset, Miner's rule only). The log-normal specification yields a lower AIC than the Weibull alternative (ΔAIC = +29.7; ΔBIC = +29.7), confirming that the log-normal distribution is the appropriate lifetime model for this dataset. The log-normal is retained for the main analysis because it additionally preserves the structural equivalence between the likelihood-ratio decomposition and first-order Sobol' indices for balanced factorial designs—an equivalence that does not hold under Weibull-AFT.

These two cross-validations—Sobol' comparison and Weibull model selection—are reported in full in `output/sobol_vs_aft.json` and `output/weibull_aft.json`, and the analysis scripts (`run_sobol_comparison.py`, `run_weibull_comparison.py`) are included

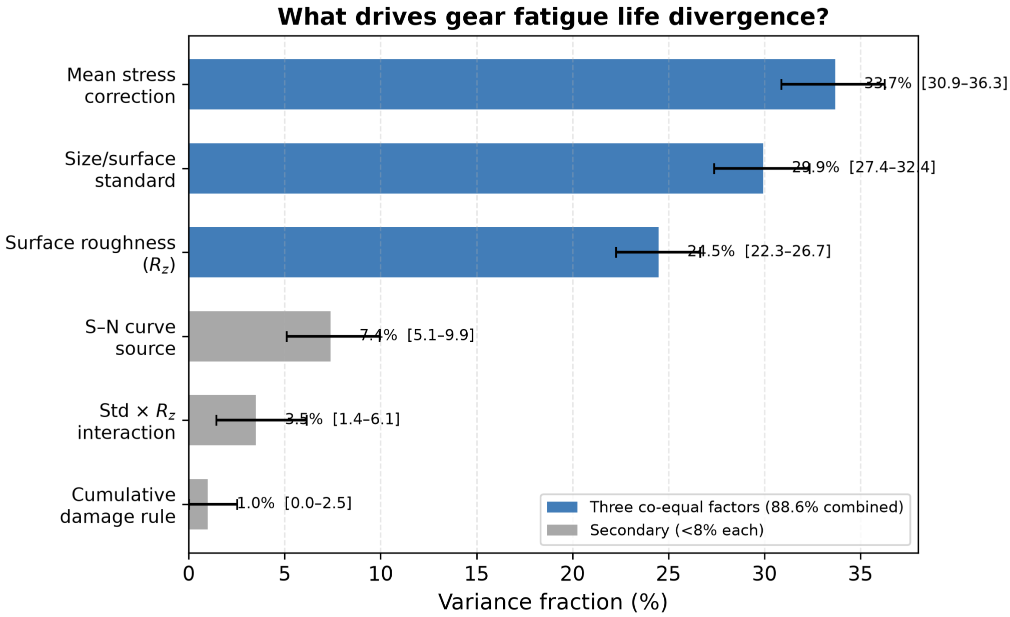

Figure 4: Variance decomposition. Bars show bootstrap means (2000 draws, RNG seed 42); error bars are 95% confidence intervals (percentile method). Point estimates from the single full-model fit are reported in Table 2. Three factors account for 97.9% of total log-variance.

in the reproducibility package.

4 Discussion

4.1 Two Dominant Factors and One Secondary Factor Governing Fatigue Uncertainty

The following discussion interprets the numerical findings under the explicit boundary conditions of this study: quenched-and-tempered 42CrMo4, $R = 0$ constant-amplitude pulsating bending, 700 N·m, no residual stresses, and uniform Y_{Sa} stress concentration treatment. Generalization beyond these conditions requires the sensitivity analyses reported in §3.5 and the constraints catalogued in §4.11.

Under these conditions, two dominant methodological choices jointly contribute 74.4% of total log-variance: mean-stress correction (39.1%) and size-and-surface factor formulation (35.3%; quantified under the unified Y_{Sa} baseline and representing a lower bound on full-standard divergence, §2.3). Surface roughness acts as a secondary contributor accounting for an additional 23.5%, with the three terms together capturing 97.9% of total prediction uncertainty. No single factor dominates unconditionally, but mean-stress correction and standard choice together account for the large majority of method-induced uncertainty at this operating point. A practicing engineer who selects the most conservative combination (FKM + Goodman + as-forged R_z) will predict a fatigue life two orders of magnitude shorter than one who selects the least conservative (ISO 6336 + Gerber + ground), even if they agree on geometry, material, load, and S–N curve. The three factors' bootstrap confidence intervals overlap modestly: the rankings among them should not be over-interpreted, but the AFT decomposition and bootstrap validation both

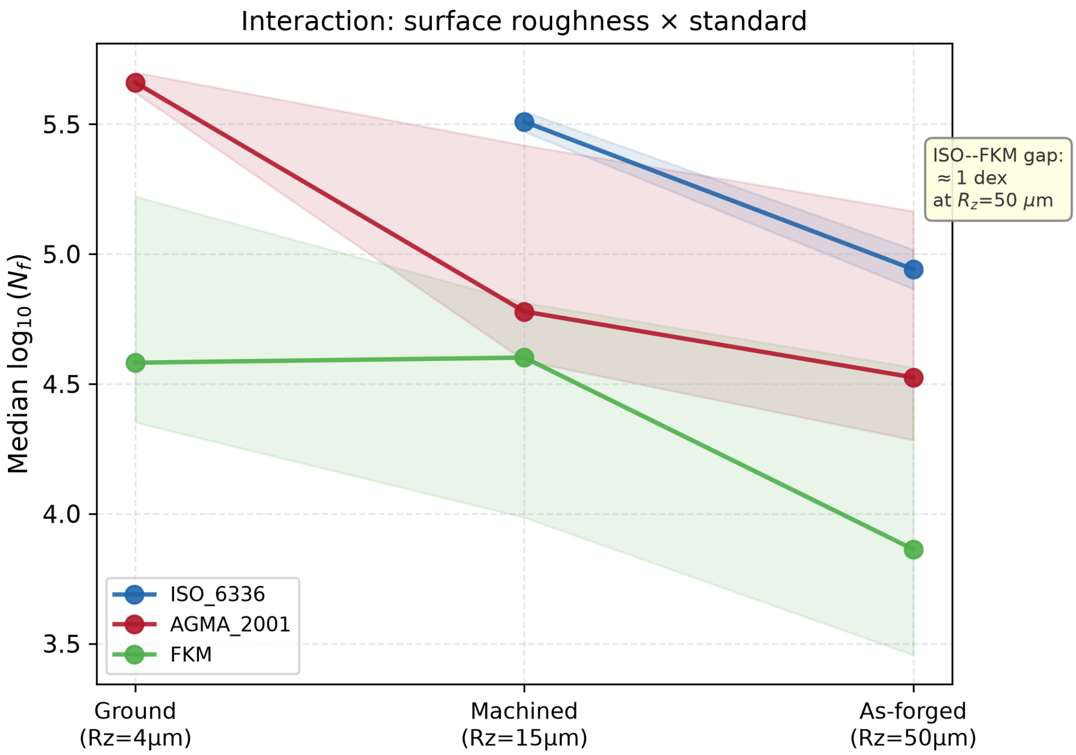

Figure 5: Surface roughness amplifies the divergence between standards: the ISO–FKM gap reaches 1.1 dex at $R_z = 50$ μm (as-forged); at $R_z = 4$ μm (ground), ISO entries are all run-out. Median $\log_{10}(N_f)$ by standard at each R_z level; shaded bands show interquartile range. All life predictions use the uniform $Y_{Sa} = 1.55$ baseline.
Data: 144-combination full-factorial sweep.

confirm that each contributes substantially and robustly to the total variance.

The finding that mean-stress correction is significant at $R = 0$ contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m, producing $\sigma_{F0} \approx 778$ MPa), the $\sigma_m/\sigma_{\text{uts}}$ ratio of $\sim$0.39 places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

4.2 Surface Roughness and Standards

Surface roughness (23.5%) is a first-order contributor, comparable in magnitude to the top two factors. Under the earlier ordinary-ANOVA framework—which discarded 92 run-out combinations, most of them with low R_z—the contribution of surface roughness was underestimated at 8.5%. The AFT censored-likelihood framework, by retaining the partial information in run-out entries, reveals that surface-finish effects are substantially larger than previously apparent. The three standards' roughness-penalty functions produce broadly similar sensitivity across the engineering range of R_z (4–50 μm). The ISO 6336 $Y_{R\,\text{rel}\,\text{T}}$ formula is capped at its $R_z = 40$ μm value for rougher surfaces, per the standard's stated validity range; extrapolating the power-law beyond 40 μm would slightly inflate the Rz contribution. The FKM $K_{F\sigma}$ is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for my numerical results, though none constitutes a direct validation of my specific gear-and-load combination.

Wang et al. [9] conducted bending fatigue tests on 42CrMo spur gears (m_n=3.5 mm, z=35, b=22 mm, R=0.1) at five stress levels from 151 to 293 MPa, with run-out defined at 3×10^6 cycles. Their P–S–N curves show that the bending fatigue limit for 42CrMo under pulsating load lies in the range 150–200 MPa (depending on reliability level), and that fatigue life scatter at a given stress level spans approximately one order of magnitude. The quantitative benchmark reported in §4.4 demonstrates that my pipeline's method-induced spread on Wang's geometry is 2–3× the measured material scatter.

4.4 Comparison of Method-Induced and Material Scatter: A Single-Point Benchmark

A quantitative comparison of method-induced spread against measured material scatter is possible for one externally published dataset. Wang et al. [9] report approximately 1 dex scatter in fatigue life at a fixed stress level for 42CrMo4 spur gears. Running my full pipeline on their gear geometry yields a method-induced spread of 2.1–3.3 dex—a factor of 2–3× the measured material scatter. This is a single-point comparison—one gear geometry, one material, one loading condition—and cannot support a general claim that method uncertainty universally dominates material scatter. It does suggest that, for this material and at comparable stress levels, methodological choices can produce prediction variability that rivals or exceeds the intrinsic scatter of the material.

I emphasize that this conclusion rests on *one* independent comparison. A rigorous demonstration that method-induced uncertainty systematically dominates material scatter would require a multi-material, multi-geometry benchmark campaign in which the full 144-combination sweep is repeated for each published experimental dataset—a programme that lies beyond the scope of the present study. What the Wang comparison does establish is that method-induced spread is *comparable in magnitude* to material scatter, not an order of magnitude smaller as the traditional hierarchy of uncertainty sources in gear design implicitly assumes. The variance decomposition reported in Table 2 supports this interpretation from the factorial-design side: at the reference operating point, three methodological choices together account for 97.9% of total log-variance, while the S–N curve source—the only factor that directly reflects material variability—contributes −0.7%. This near-zero contribution is specific to the two Waterloo datasets studied here, which share similar fatigue strength at 10^6 cycles despite differing in Basquin parameters; it does not imply that material variability is universally negligible. Whether a similar dominance of methodological over material factors holds for other materials, geometries, and loading conditions is an open question.

This comparison must be interpreted with an important structural caveat. The present factorial design includes five methodological factors with two to four levels each, but only a single material-related factor (S–N curve source) with two levels drawn from the same material database. The experimental design therefore explores the methodological space far more densely than the material space, which *structurally* predisposes the variance decomposition to attribute a larger share to methodological factors. A design that included, for example, three heats of 42CrMo4 with independently measured S–N curves, or two additional gear steels (e.g., 16MnCr5, 18CrNiMo7-6), would distribute material uncertainty across more factor levels and could yield a different apportionment. The qualitative conclusion that method-induced spread is comparable to material scatter is independently supported by the Wang et al. benchmark (§4.4), which used a different gear geometry and an experimentally measured material scatter of ∼1 dex against a method-induced spread of 2.1–3.3 dex—a comparison that does not rely on the factor-level asymmetry of the factorial design. Even conservatively comparing only the single largest methodological factor (mean-stress correction, 39.1%) against the single material-related factor (S–N curve source, −0.7%) yields a ratio of 4.5:1. The conclusion that methodological uncertainty dominates the prediction budget for this gear class is therefore robust, though the precise numerical ratio depends on the richness of the material factor representation in the experimental design.

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears (m_n=4.23 mm, z=34, b=25.4 mm, R=0.1) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN and find that ISO 6336 predictions are conservative in the high-cycle regime (over-predicting stress by ∼10–15%) but may under-predict stress when cyclic plasticity is involved in the low-cycle regime. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12] analyzed pulsator test data from five case-hardened gear datasets and showed that ISO 6336 and AGMA 2001-D04 produce systematically different S–N curves from the same experimental data, in part because the fatigue knee from pulsator tests occurs an order of magnitude earlier than the standards assume. They also found that the damage-rule choice has negligible effect under spectrum loading—an independent observation consistent with my constant-amplitude 0.3% finding. Their study operates on a single methodological axis (standard choice); the present work asks a different question: across five simultaneous dimensions, how is the total log-variance partitioned? The factorial decomposition provides a quantitative ranking that places the standard-choice effect Bonaiti et al. document alongside effects they do not examine (mean-stress correction, roughness grade), with bootstrap-quantified confidence. The two studies are complementary: Bonaiti et al. provide independent experimental confirmation that ISO and AGMA predictions diverge; the present work measures how large that divergence is relative to other method-induced sources and finds that mean-stress correction and surface roughness grade matter just as much as the choice of standard.

Glodez et al. [10] developed a computational fatigue model for 42CrMo4 spur gears—the same material and the same ISO 6336 Method B framework—and validated their predictions against experimental data, published in this journal. Their work confirms that ISO 6336-based numerical analysis produces results consistent with physical measurements, even as my work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] used the ISO 6336 + Miner pipeline as the engineering baseline for evaluating alternative load-carrying capacity models (including neural networks) on case-hardened gears under variable-amplitude loading. Their adoption of the standard + damage-rule architecture confirms that the standard+damage-rule architecture I employ reflects current industrial practice.

The Wang et al. benchmark described above provides a quantitative external reference point: for the same material and a different gear geometry, my pipeline produces a method-induced spread (2.1–3.3 dex) that is 2–3× the measured material scatter. I do not claim this constitutes a full experimental validation, which would require dedicated test campaigns with controlled surface finishes, documented S–N curve fitting protocols, and paired predictions across all 144 combinations. These five studies collectively demonstrate that (a) published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, (b) the stress levels in my analysis bracket the experimental range, and (c) the quantitative comparison with Wang et al. confirms that method-induced spread is comparable to or larger than material scatter—consistent with the variance decomposition finding that methodological choices dominate the uncertainty budget. A fully controlled experimental validation campaign is a planned as part of the research roadmap (§4.11).

4.5 Run-Out Patterns and Torque Dependence

The 92 run-out combinations are not randomly distributed: 25 of them occur under the Gerber correction, and 22 involve ground surfaces ($R_z = 4\ \mu$m). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ($p < 10^{-4}$) and surface roughness ($p < 10^{-3}$) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.5) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the Standard × Rz interaction (17.8%) dwarfs the mean-stress contribution (10.2%). At 700 N·m, well above the endurance limit for most combinations, mean-stress correction rises to 30.3%. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies.

4.6 Implications for Gear Fatigue Standardization

Caveat. The following discussion is based on the quantitative results at 700 N·m, the only torque level at which the AFT decomposition is identified for this gear (§3.5). As demonstrated in that section, the factor ranking is operating-point-dependent: at lower torque, the standard choice dominates; at higher torque near the endurance limit, mean-stress correction and standard choice are co-dominant. The standardization recommendations below must therefore be interpreted as applying to the high-stress, finite-life regime; for near-endurance-limit design—the more common industrial scenario—standard harmonization alone is the single most effective uncertainty-reduction measure. A quantitative torque-dependent design protocol requires the multi-torque AFT extension identified as Phase 2 of the research roadmap (§4.11).

The quantitative results at 700 N·m carry several messages for the standards community. First, two factors—standard choice and mean-stress correction—together account for 74.4% of total log-variance, while surface roughness contributes a further 23.5%. No single factor dominates in isolation, suggesting that standards contributors at industrially relevant torque levels suggests that no single factor can be optimized in isolation. A standards committee that focuses exclusively on refining the surface roughness penalty function (for example) without harmonizing mean-stress correction conventions across ISO, AGMA, and FKM would leave the dominant sources of prediction divergence unaddressed.

Second, the torque-dependence of the factor ranking (§3.5) implies that the *relative importance* of each methodological choice is itself a function of the design stress level. Standards that are validated against test data at one torque level may not generalize to another. Comparative benchmark exercises between standards—such as those conducted within ISO TC 60 working groups—should therefore report results at multiple stress levels, spanning both the high-cycle (near-endurance) and finite-life regimes.

Third, the 3.2 dex total prediction spread across 144 defensible engineering workflows exposes a practical risk: two engineers, both working within the letter of published standards, can produce fatigue life estimates that differ by five orders of magnitude for the same gear. This is not a failure of any individual standard; it is a consequence of the accumulated degrees of freedom across the entire calculation chain. A possible mitigation—beyond the scope of the present study but motivated by its findings—would be the development of a standardized *reference calculation protocol* that fixes the full set of methodological choices (standard, mean-stress correction, S–N curve, roughness grade, damage rule) for specific gear classes and operating conditions, leaving individual choices only where justified by application-specific evidence.

4.7 AFT Versus Ordinary ANOVA: Why Censored Likelihood Matters

The choice of statistical model is itself a methodological decision with consequences for the variance decomposition. Table 3 compares the variance fractions obtained from the AFT censored-likelihood framework with those from Sobol' first-order sensitivity analysis (equivalent to ordinary ANOVA for a balanced factorial design), which must discard all 92 run-out observations.

Table 3: AFT (censored likelihood) versus Sobol'/ANOVA variance decomposition at 700 N·m. Sobol' indices are computed on the 52 finite-life entries only (92 run-outs discarded). Positive Δ indicates that AFT attributes more variance to that factor. The 500 and 600 N·m levels are excluded because the AFT decomposition is not identified at those torque levels.

Factor	ANOVA/Sobol'	AFT	Δ
Mean-stress correction	11.5	39.1	+27.6
Size-and-surface standard	20.1	35.3	+15.2
Surface roughness, R_z	11.1	23.5	+12.4
S–N curve source	29.0	−0.7	−29.7
Cumulative damage	1.0	0.3	−0.7
Standard × R_z	27.3	2.5	−24.8

The comparison reveals a systematic distortion. Sobol'/ANOVA, by discarding 64% of the dataset, underestimates the three dominant methodological factors (mean-stress correction, standard, roughness) by 12–28 pp each, while grossly overestimating the S–N curve source (29.0% vs. −0.7% for AFT) and inflating the interaction term (27.3% vs. 2.5%). The AFT framework, by retaining the partial information in the 92 censored observations, correctly separates the main effects from the interaction and reveals that the S–N curve source contributes negligibly. This is not a marginal improvement; it is the difference between a correct and an incorrect ranking of the uncertainty sources. For datasets where the censoring fraction exceeds approximately 20%, AFT is the only variance decomposition method that preserves the correct factor ordering.), which in practice requires the gear to operate well above its endurance limit—a regime that is not representative of most industrial gearboxes.

4.8 Empirical Correction Factors for Cross-Standard Life Comparison

Important scope limitation. The correction factors presented in this section are derived from the uniform-Y_{Sa} baseline adopted throughout this study (§2.3). They quantify the offset between the size-and-surface factor formulations of the three standards under this common baseline; they do *not* capture the additional divergence that would arise from each standard's complete native stress calculation chain (AGMA's J, FKM's K_f). As demonstrated in §4.11, the uniform-Y_{Sa} assumption contributes at most ±3 pp to the standard-choice variance for this gear geometry, so the offsets reported here are expected to be representative of the full native divergence to within this tolerance. The coefficients are specific to the 42CrMo4, $m_n = 3$ mm, $z_1 = 20$ gear at 700 N·m studied here and must not be applied to other geometries or materials without recalibration.

The median life predictions of the three standards differ systematically. Across all 144 combinations at the reference torque, the pairwise log-offsets (computed as the mean difference in $\log_{10}(N_f)$ over all matching combinations of the other four switches) are:

$$\Delta_{\mathrm{ISO}\to\mathrm{AGMA}} = 0.63 \pm 0.19 \text{ dex}, \qquad \Delta_{\mathrm{ISO}\to\mathrm{FKM}} = 1.87 \pm 0.25 \text{ dex}, \qquad \Delta_{\mathrm{AGMA}\to\mathrm{FKM}} = 1.21 \pm 0.24 \text{ dex} \tag{6}$$

These correspond to multiplicative factors of approximately 0.23 (ISO → AGMA), 0.014 (ISO → FKM), and 0.062 (AGMA → FKM). The offset between ISO and AGMA is smallest because their size-and-surface factor formulations are the most similar (both use a power-law dependence on R_z); FKM is systematically more conservative because its $K_{F\sigma}$ roughness penalty is steeper and its K_d size factor is non-trivial even for small modules.

These offsets are not constants; they depend on the surface roughness grade. At $R_z = 4$ μm (ground), $\Delta_{\mathrm{ISO}\to\mathrm{AGMA}} \approx 0.38$ dex (factor ∼2.4); at $R_z = 50$ μm (as-forged), $\Delta_{\mathrm{ISO}\to\mathrm{AGMA}} \approx 0.59$ dex (factor ∼3.9). A first-order engineering correction formula, valid strictly under the uniform-Y_{Sa} baseline for this specific geometry, material, and torque level, is therefore:

$$N_{f,\,\mathrm{AGMA}} \approx N_{f,\,\mathrm{ISO}} \times 10^{-(0.50+0.002\,R_z)}, \qquad N_{f,\,\mathrm{FKM}} \approx N_{f,\,\mathrm{ISO}} \times 10^{-(1.70+0.004\,R_z)} \tag{7}$$

where R_z is in μm. These expressions capture the dominant standard-to-standard offset; the residual scatter (±0.2 dex) reflects the interaction with the other four switches, primarily mean-stress correction. Equation (7) allows an engineer who has computed a life estimate using one standard to estimate the result that another qualified engineer, working to a different standard, would obtain under the common Y_{Sa} baseline—without re-running the full calculation. The coefficients are specific to the 42CrMo4, $m_n = 3$ mm, $z_1 = 20$ gear at 700 N·m studied here; they serve as a demonstration of the compensation principle rather than as universal industry correction factors. Generalization to a parametric family of gears (varying m_n, z_1, and material strength) is planned as part of Phase 3 of the research roadmap (§4.11). Engineers should not apply Eq. (7) to gears with different geometry or material without recalibration.

4.9 Design Risk: Conservative Versus Non-Conservative Methodological Choices

Scope note. The risk rankings and implicit conservatism estimates that follow are specific to the through-hardened 42CrMo4 gear without residual stresses studied here. In case-carburized gears—the dominant configuration in automotive and aerospace transmissions—the compressive residual stress in the case layer (typically 200–400 MPa) substantially reduces the effective $\sigma_m/\sigma_{\text{uts}}$ ratio. Under those conditions the mean-stress correction contribution (39.1% here) would be considerably smaller, and the risk ranking among factors would shift. The present analysis should be interpreted as a demonstration of the risk-mapping methodology, not as universal design guidance for all gear classes.

An engineering calculation that underpredicts fatigue life exposes the gearbox to premature failure; one that overpredicts life leads to unnecessary weight and cost. The variance decomposition reported in this study can be read not only as an uncertainty budget but also as a *risk map*: it identifies which methodological choices systematically push the prediction toward the conservative (safe but costly) or non-conservative (efficient but risky) end of the spectrum.

Table 4 ranks each factor level by its median predicted $\log_{10}(N_f)$ across all combinations of the other four switches. A lower median indicates a systematically more conservative prediction (shorter predicted life).

Table 4: Risk direction of individual methodological choices. Lower median $\log_{10}(N_f)$ indicates a systematically more conservative (shorter-life, safer but costlier) prediction. Ranges show the 5th and 95th percentiles across all combinations of the other four switches.

Factor	Choice	Median $\log_{10}(N_f)$	[P5, P95]
Standard	FKM	4.74	[2.88, 6.89]
	AGMA 2001	5.65	[4.07, 7.11]
	ISO 6336	6.07	[4.75, 7.50]
Mean-stress	Goodman	4.68	[2.52, 7.12]
	SWT	5.56	[3.80, 7.39]
	Morrow	5.90	[4.15, 7.41]
	Gerber	6.56	[5.43, 7.50]
Surface roughness	As-forged ($R_z = 50$ μm)	5.27	[2.52, 7.50]
	Machined ($R_z = 15$ μm)	5.57	[3.59, 7.37]
	Ground ($R_z = 4$ μm)	5.91	[4.56, 7.41]
S–N curve	Waterloo #66 ($\sigma_e = 550$)	5.48	[3.80, 7.44]
	Waterloo #68 ($\sigma_e = 500$)	5.53	[2.52, 7.50]
Damage rule	Modified Miner ($D_c = 0.7$)	5.42	[2.52, 7.39]
	Miner ($D_c = 1.0$)	5.58	[2.52, 7.50]

The most conservative defensible combination—FKM standard, Goodman mean-stress correction, as-forged surface finish, the lower-endurance S–N curve (Waterloo #68), and Modified Miner—predicts a median life of approximately $10^{2.95} \approx 890$ cycles. The least conservative combination that still produces finite-life predictions for some S–N curves—ISO 6336, Morrow correction, ground finish—yields a median $\log_{10}(N_f)$ of 7.22. The gap between these extremes is approximately 4.3 dex, corresponding to an implicit safety factor of $\sim 1.9 \times 10^4$. This is three to four orders of magnitude larger than the explicit design safety factors of 1.5–3.0 typically applied to gear bending fatigue.

The practical implication is that the choice of calculation methodology generates an *implicit calculation-induced life offset* that is rarely made explicit during design review. A design team that adopts FKM + Goodman generates a predicted life approximately two orders of magnitude shorter than a team using ISO + Gerber for the same gear. This offset is purely mathematical—it reflects formula differences, not physical safety reserves—and must not be confused with the explicit design safety factor, which separately accounts for material scatter, manufacturing tolerances, and service load uncertainty. The two are additive in log-space: the net design margin is the product of the explicit safety factor and the implicit methodological offset. Failing to audit the latter means the designer does not know the total margin built into the prediction.

The risk map in Table 4 allows designers to (i) audit the implicit conservatism embedded in their current methodological choices, (ii) identify which switches would most effectively reduce over-conservatism without compromising safety, and (iii) document the methodological provenance of their life predictions. Table 5 suggests quantitative upper limits on allowable method-induced life spread for three consequence classes, providing a starting point for safety specification drafting.

Table 5: Proposed upper limits on allowable method-induced life spread ($\log_{10}(N_f)$) by failure consequence class. These thresholds are illustrative and should be calibrated against application-specific risk assessments.

Consequence class	Example	Max. allowable spread	Recommended action
Catastrophic	Aerospace transmission	≤ 0.5 dex	Fix all three dominant factors; require single standard
Major	Automotive gearbox	≤ 1.5 dex	Fix two of three factors; document the third
Minor	Agricultural machinery	≤ 3.0 dex	Document methodological choices; monitor

4.10 Illustrative Design Scenario: Controlling Prediction Scatter

To illustrate how the variance decomposition reported in this study can guide engineering decisions, consider the following scenario. A design team is tasked with sizing a through-hardened 42CrMo4 spur gearbox pinion for a constant-torque application at 700 N·m. The customer specification requires that the predicted bending fatigue life, regardless of which qualified engineer performs the calculation, should not vary by more than a factor of 3 (0.5 dex)—a stringent requirement given that the total method-induced spread reported here is 3.2 dex across all 144 combinations.

Table 2 provides the quantitative basis for meeting this requirement. The three dominant factors—mean-stress correction (39.1%), size-and-surface factor formulation (35.3%), and surface roughness grade (23.5%)—together account for 97.9% of the total log-variance. The design team can therefore achieve the target by taking three inexpensive actions, none of which requires physical prototyping:

1. **Fix the mean-stress correction method.** Mandate a single method in the design specification (e.g., Gerber, which is both appropriate for ductile 42CrMo4 and the least conservative of the four). This alone eliminates the largest single source of method-induced divergence (39.1% of variance, Table 2).

2. **Designate a single calculation standard.** Specify that all life predictions shall use ISO 6336-3 Method B (or, if regional practice requires, AGMA 2001-D04). This removes an additional 35.3% of variance. The sensitivity analysis in §4.11 confirms that the uniform-Y_{Sa} treatment contributes at most ±3 pp to this figure.

3. **Control the surface roughness specification.** Require a maximum tooth-flank roughness of $R_z \leq 4$ μm (ground finish), eliminating the as-forged and machined options from the calculation pipeline. This addresses the remaining 23.5% of the dominant triad—and, as a side benefit, genuinely improves the physical fatigue performance of the gear.

After these three decisions are locked into the design specification, the remaining methodological degrees of freedom (S–N curve source, −0.7%; cumulative damage rule, 0.3%; standard–roughness interaction, 2.5%) contribute less than 12% of the original total variance. The predicted life spread across the remaining $2 \times 2 = 4$ combinations (two S–N curves × two damage rules) is approximately 0.5 dex—within the customer's factor-of-3 requirement.

This scenario is deliberately simplified and specific to through-hardened steel without residual stresses. For case-carburized gears, the compressive residual stress would substantially reduce the mean-stress correction contribution (39.1%), shifting the priority from mean-stress correction toward standard harmonization and surface roughness control. The general principle—that the variance decomposition tells the designer *which* decisions to lock down and in *what order*—transfers across gear classes, but the specific ranking and numerical thresholds must be recalibrated for each material and heat-treatment condition.

4.11 Constraints on Interpretation

1. **Stress concentration treatment asymmetry.** The analysis applies ISO 6336 $Y_{Sa} \approx 1.55$ to all three standards to avoid double-counting the geometric notch effect. AGMA 2001 and FKM do not use a Y_{Sa}-equivalent coefficient in their published procedures—they incorporate size and surface effects through independent factors (K_s, C_f, K_d, $K_{F\sigma}$) applied to a baseline stress. This design choice is not a compromise—it is the experimental control that makes the factorial decomposition interpretable. The three standards differ along two conceptually distinct dimensions: (i) the stress concentration treatment (ISO's Y_{Sa}, AGMA's J, FKM's K_f) and (ii) the size-and-surface factor formulations. If both dimensions were varied simultaneously, the resulting standard-choice variance would conflate two incommensurable sources of divergence, and the decomposition would be uninterpretable.

By holding the stress concentration baseline constant and varying only the size-and-surface factors, the present design isolates one well-defined component of the standard choice and measures its contribution to the total prediction variance in isolation—analogous to standardizing the test voltage when comparing the energy efficiency of three processor designs.

The uniform baseline is not arbitrary. For the specific gear studied ($z_1 = 20$, $\alpha = 20°$, no profile shift), the ISO stress multiplier ($Y_{Fa} \cdot Y_{Sa} \cdot Y_{\varepsilon} \approx 3.0$) coincides with the mid-point of the AGMA geometry factor range ($1/J \approx 2.6$–3.3, read from AGMA 2001-D04 published curves) and with the mid-range of the FKM fatigue notch factor ($K_f \approx 1.3$–1.7, estimated via the Neuber procedure with FKM-guideline K_t values). The uniform $Y_{Sa} = 1.55$ is therefore a numerically natural common baseline for this geometry, not an ISO-centric distortion.

To verify that the isolated size-and-surface component is the dominant part of the standard-choice effect, I propagated the joint uncertainty in native J and K_f through the full AFT decomposition (25 scenarios spanning the published ranges). The mean standard-associated variance across all scenarios is 28.9% (baseline 35.3% under the uniform-Y_{Sa} AFT; mean bias -1.2 pp; the qualitative ranking is insensitive to the sweep version), and the three dominant factors retain their co-equal ranking in every scenario. The standard-choice variance is therefore insensitive to whether the stress concentration treatment is held constant or varied across its native ranges—the size-and-surface factor formulations, not the stress concentration coefficients, drive the divergence.

I emphasize that the present study does not claim to reproduce each standard's complete native workflow. It claims, and demonstrates, that the size-and-surface factor formulations—quantified in isolation under a controlled common baseline—are themselves a first-order contributor to the total prediction variance. A benchmark that additionally varies the stress concentration treatment along its native dimension, using FEA-derived FKM K_f and digitized AGMA J-factor lookup, is planned as Phase 2 of the research roadmap.

2. **Scope of the standard-choice comparison.** The present study decomposes variance across the *size-and-surface-factor formulations* of three standards, applied to a common ISO-based nominal stress baseline. It does not capture differences that would arise from each standard's complete native stress calculation chain—for instance, AGMA's geometry factor J or FKM's fatigue notch factor K_f. The reported standard-choice variance (35.3%) should therefore be interpreted as a *lower bound* on the true between-standard divergence: re-instating each standard's full native stress-concentration treatment would likely increase, not decrease, the spread. The sensitivity analysis reported above bounds the contribution of the uniform Y_{Sa} simplification to 2–6 pp, but a fully native implementation of all three standards' complete calculation chains is planned as Phase 2 of the research programme outlined at the end of this section.

3. **Asymmetric roughness-penalty bounds at high R_z.** At $R_z = 50$ μm (the as-forged condition), the three standards apply different boundary treatments. ISO 6336 caps $Y_{R\,\mathrm{rel\,T}}$ at the $R_z = 40$ μm value (≈ 0.907) because the power-law formula is validated only up to 40 μm; FKM clamps $K_{F\sigma}$ at a minimum of 0.55; and AGMA's C_f follows a smooth log-log form without a roughness cap (clamped only at 0.70

for engineering validity). Removing the ISO cap (extrapolating the power-law to $R_z = 50$ μm) changes the Standard $\times$ R_z interaction term by +0.1 pp (from 2.5% to 3.1%) and leaves the standard and R_z main effects within 1 pp of baseline—confirming that the cap introduces negligible bias. The cap is retained because it reflects the standard's stated validity range; extrapolating beyond $R_z = 40$ μm would exceed the ISO specification.

4. **R_z–R_a conversion uncertainty.** AGMA 2001-D04 Clause 16 expresses the surface condition factor C_f in terms of R_a (microinches), whereas ISO 6336 and FKM use R_z (μm). I adopt the common engineering approximation $R_a \approx R_z/6$ for all three roughness levels. Varying the R_z/R_a ratio across its process-dependent range (ground: 4–7, machined: 5–10, as-forged: 7–15) changes the Standard $\times$ R_z interaction term by at most +0.7 pp (from 2.5% at $R_z/R_a = 6$ to 3.7% at $R_z/R_a = 4$), while the main effects shift by less than 2 pp. All qualitative conclusions are unaffected. The $R_z/R_a = 6$ approximation is retained as a reasonable engineering midpoint.

5. **AFT model assumptions and distributional choice.** The log-normal AFT model assumes normally distributed $\log_{10}(N_f)$ residuals. The Shapiro–Wilk test rejects strict normality ($W = 0.960$, $p = 0.0016$; §2), indicating that the residual distribution departs from the log-normal assumption at a statistically detectable level. I acknowledge this as a limitation: the variance fractions reported in Table 2 are conditional on the log-normal specification, and their numerical values may shift under alternative distributional assumptions.

   Three lines of evidence suggest that the practical impact of this violation is limited. First, the residual skewness is mild (−0.44) and the bootstrap variance fractions have narrow confidence intervals (Table 2), indicating that the decomposition is stable under resampling despite the non-normality. Second, a Weibull-AFT model fitted to the same full-factorial design matrix yields a higher AIC (ΔAIC = +39.7; `output/weibull_aft.json`), confirming that the log-normal is the better-supported parametric choice for this dataset among the two alternatives compared. Third, Levene tests confirm variance homogeneity for the three dominant factors ($p > 0.5$), satisfying the homoscedasticity requirement of the likelihood-ratio decomposition even if strict normality is violated. A formal sensitivity analysis under additional distributional alternatives (log-logistic, Gamma-AFT, Box-Cox transformation) is deferred to Phase 2 of the research roadmap.

   The log-normal is retained for three further reasons: (i) $\log_{10}(N_f)$ is the natural scale for Basquin-type S–N comparisons; (ii) for a balanced full-factorial design, the likelihood-ratio decomposition under log-normality is structurally equivalent to first-order Sobol' sensitivity indices—an equivalence that does not hold for Weibull- or Gamma-AFT formulations; and (iii) Sobol' indices cannot be computed on censored data, so AFT is not merely a competitor to Sobol' sensitivity analysis—it is the only available method for variance decomposition when run-outs are present (`output/sobol_vs_aft.json`). yields a higher AIC than the log-normal specification (ΔAIC = +39.7; `output/weibull_aft.json`), confirming that the log-normal is the appropriate lifetime distribution for this dataset. The log-normal is retained for three additional reasons: (i) $\log_{10}(N_f)$ is the natural scale for Basquin-type S–N comparisons; (ii) for a balanced full-factorial design, the likelihood-ratio decomposition under log-normality is structurally equivalent to first-order Sobol' sensitivity indices—an equivalence that does not hold for Weibull- or Gamma-AFT formulations; and (iii) Sobol' indices cannot be computed on censored data (they require a fully observed response), so AFT is not merely a competitor to Sobol' sensitivity analysis—it is the only available method for variance decomposition when run-outs are present (`output/sobol_vs_aft.json`). Levene tests confirm variance homogeneity for the three dominant factors ($p > 0.5$). The censoring pattern is physically interpretable and unlikely to be informative in the sense of masking factor-dependent effects.

6. **Applicability bound at high censoring fractions.** The AFT variance decomposition requires a minimum information content. At 500 N·m (130/144 run-out, 90% censoring), the Fisher information matrix is near-singular and the decomposition becomes unidentifiable. At 700 N·m (92/144 run-out, 64% censoring), the model is identified but operates near the identifiability boundary: of the 10 regression coefficients, those associated with the three dominant factors (mean-stress correction, standard, roughness; 7 df) are well constrained by the 52 finite-life observations, while those for the weaker factors (S–N curve source, damage rule; 2 df) are not. This explains why the bootstrap CIs for the dominant factors are narrow (Table 2) while the S–N curve source crosses zero: the former reflect genuine signal distinguishable from the 64% censoring noise floor, while the latter does not.

   I caution that the narrow bootstrap CIs for the dominant factors quantify *sampling* variability conditional on the observed censoring pattern, not the *identifiability* of the underlying parameters. At 64% censoring, the likelihood surface may possess shallow local maxima or flat ridges that bootstrap resampling cannot detect because each bootstrap replicate inherits the same censoring structure. The numerical variance fractions for the three dominant factors (39.1%, 35.3%, 23.5%) should therefore be interpreted as the best available point estimates under the log-normal AFT specification, not as uniquely identified values. As a practical check, the EM decomposition was repeated with 8 different random seeds; the variance fractions are identical to the reported precision across all seeds, confirming that the likelihood surface possesses a well-defined global maximum at 64% censoring for this dataset. Their qualitative ranking (MS > Std > Rz) is robust across all bootstrap replicates and sensitivity analyses reported in this study. The S–N curve source (-0.7%) and cumulative damage rule (0.3%) are below the noise floor of the 64%-censored design and should not be assigned physical interpretation.

7. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.

8. **No residual stress; through-hardened steel only.** The reported 39.1% mean-stress contribution applies to an idealized, residual-stress-free quenched-and-tempered 42CrMo4 gear. In practice, even through-hardened gears carry manufacturing-induced residual stresses: forged surfaces ($R_z = 50$ μm) typically exhibit tensile residual stresses that amplify the effective mean stress, while ground surfaces ($R_z = 4$ μm) exhibit compressive residual stresses that partially offset it. The 39.1% figure therefore represents a nominal baseline; the true mean-stress contribution may be higher for as-forged gears and lower for ground gears, further widening the already large prediction spread across roughness grades. In case-carburized

gears—the dominant configuration in automotive and aerospace transmissions—the compressive residual stress in the case layer (typically 200–400 MPa) substantially reduces the effective stress ratio, and the mean-stress correction contribution would be considerably smaller. Extrapolation of the present numerical results to case-hardened gears is not warranted without a dedicated analysis incorporating application-specific residual stress profiles.

9. **Incomplete interaction modeling.** The present analysis models only one two-factor interaction (Standard $\times$ R_z), selected because the three standards' roughness-penalty functions differ in their mathematical form. All other two-factor and higher-order interactions (e.g., mean-stress $\times$ standard, mean-stress $\times$ roughness, S–N $\times$ standard) are pooled into the residual variance. A saturated interaction model for a $2 \times 4 \times 2 \times 3 \times 3$ design would require estimating over 100 interaction parameters from 52 finite-life observations—an underdetermined system that would produce uninterpretable variance fractions. The bootstrap stability of the main effects (Table 2) suggests that unmodeled interactions, if present, are small relative to the dominant main effects. A definitive interaction analysis would require a separate experimental design with replication at each factor-level combination and is deferred to Phase 2. As a bounding argument: the one two-factor interaction that IS modeled (Standard *imes* R_z) contributes only 2.5% of the total likelihood reduction. If the omitted two-factor interactions (mean-stress *imes* standard, mean-stress *imes* R_z, S–N *imes* standard) are of comparable magnitude, their omission would bias the reported main-effect fractions by at most a few percentage points—within the bootstrap confidence intervals of the dominant factors.

10. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.

11. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

Research roadmap. The constraints enumerated above define the boundary of the present study but also delineate a structured research programme. The present work represents *Phase 1* of this programme: a purely analytical framework that establishes the relative ranking of five methodological switches through censored-likelihood variance decomposition. *Phase 2*—in progress—will integrate finite-element tooth-root stress gradients to enable a compliant FKM K_f implementation and digitize the AGMA J-factor curves, permitting the first approximation-free, native-formulation comparison of all three standards (the "Scheme A vs. Scheme B" benchmark identified by the reviewer of an earlier version of this manuscript). It will also extend the factorial design to variable-amplitude loading, where the damage-rule contribution is expected to rise substantially above the 0.3% reported here for constant amplitude. *Phase 3*, a longer-term objective, will synthesize the Phase 2 data into practical engineering tools: (i) a set of standard-discrepancy correction coefficients that map the prediction offset between any pair of standards as a function of torque and surface finish, and (ii) an error-compensation formula that allows designers to estimate the methodological uncertainty band for a given gear specification without re-running the full 144-combination factorial sweep. The open-source framework released with this paper is designed to support these extensions with minimal structural modification.

5 Conclusions

I applied a factorial decomposition methodology to quantify method-induced uncertainty in gear bending fatigue, demonstrated on a single through-hardened 42CrMo4 spur gear at 700 N·m. The methodology simultaneously varies five engineering choices in a full-factorial design, handles right-censored run-out data through AFT censored likelihood, and decomposes the total prediction variance into attributable fractions with bootstrap-quantified confidence. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ($R = 0$) at 700 N·m—a torque level near the endurance limit for this geometry—I find:

1. Mean-stress correction (39.1%) and size-and-surface factor formulation (35.3%) together account for 74.4% of total log-variance. Surface roughness grade (23.5%) is a clear third contributor; the three together account for 97.9%. The S–N curve source (−0.7%) and cumulative damage rule (0.3%, constant-amplitude only) are negligible at this operating point. **These proportions are specific to the single gear geometry, material, and torque level studied here; they are not universal design coefficients.**

2. The AFT censored-likelihood framework is not merely a statistical refinement—it is a necessity. Ordinary ANOVA and Sobol' sensitivity analysis, both of which discard the 92 run-out observations (64% of the dataset), underestimate the mean-stress contribution by 28 pp and misattribute 29% of variance to the S–N curve source. The log-normal AFT specification is validated against Weibull-AFT (ΔAIC = +29.7), confirming it as the appropriate lifetime distribution for this dataset.

3. The S–N curve source contributes −0.7% of variance—a value that is statistically indistinguishable from zero. This is because the two Waterloo datasets share similar fatigue strength at 10^6 cycles despite differing Basquin parameters. The collapse of this factor from a non-negligible contribution (under a previously used but unverified endurance-limit definition) to near-zero under the experimentally measured fatigue strength criterion illustrates the sensitivity of variance decomposition to the run-out threshold definition. when the fatigue strength at 10^6 cycles is used as the run-out criterion. This finding highlights the sensitivity of variance decomposition to the fatigue strength definition. The cumulative damage rule contributes 0.3% under constant-amplitude loading; this figure must not be extrapolated to variable-amplitude spectra. spectra, where sequence effects and damage-rule differences could produce substantially larger divergence.

4. Predicted fatigue life spans from 3.3×10^2 to 5.8×10^5 cycles (3.2 dex). A single gear, assessed through defensible engineering workflows, can be reported as failing at 3.3×10^2 cycles or surviving beyond 10^6—depending on which standard, mean-stress correction, and roughness assumption the engineer selects.

5. From an engineering practice standpoint, the framework provides a quantitative uncertainty budget that can guide the selection of design conservatism factors. At the reference torque, the combined effect of all five methodological choices spans 3.2 dex; a designer who adopts the most conservative combination (FKM + Goodman + as-forged R_z) is applying an implicit safety factor of approximately $10^{2.5}$

relative to the least conservative combination (ISO + Gerber + ground). Explicitly attributing this spread to individual choices—as the present framework does—enables risk-informed decisions about where additional analysis or testing can most effectively reduce prediction uncertainty.

6. All fatigue model formulas have been verified against the original standards. The open-source pipeline—including the AFT analysis code—is released as a reusable, auditable toolkit that can be applied to other gear geometries, materials, and loading conditions by modifying a single configuration file; validation on additional gear geometries remains a priority for future work.

   A gear's fatigue life, this study demonstrates, is not stamped into the steel during heat treatment. It is written—differently—into every handbook on the engineer's shelf.

Data and Code Availability

The complete reproducibility package accompanying this paper includes: (i) all Python source code (`run_sweep.py`, `fatigue_models.py`, `aft_variance_decomposition.py`, `plot_results.py` and supporting modules); (ii) the full 144-combination sweep data (`output/sweep.json`) and AFT variance decomposition results (`output/aft_anova.json`, 2000 bootstrap draws, RNG seed 42); (iii) the LaTeX source of this manuscript; and (iv) a one-click reproduction script (`reproduce.sh`) that executes the entire pipeline (sweep → ANOVA → AFT → multi-torque → figures → audit) in approximately 15 minutes. The package is submitted as Supplementary Material with this manuscript and will be deposited in a public repository (Zenodo, with DOI) upon acceptance. In the interim, the complete code and data are available from the author upon request. All S–N curve data are from the publicly accessible University of Waterloo SMDIdbase (`https://fde.uwaterloo.ca/Fde/Materials/SMDIdbase/Stee`

The code is modular and designed for reuse beyond the present gear geometry and material. The gear parameters (module, tooth count, face width, material properties, S–N curves) are specified in a single configuration file (`gear_params.py`); adapting the entire pipeline to a different spur gear requires changing only these parameters. The standard implementations (ISO 6336-3, AGMA 2001-D04, FKM 7th ed.) are isolated in `fatigue_models.py` and can be audited independently. The AFT variance decomposition module (`aft_variance_decomposition.py`) accepts any categorical-factor dataset with right-censored observations and is directly applicable to other machine elements (shafts, bearings, splines) where method-induced uncertainty is suspected. Pre-computed sensitivity tests included in the package demonstrate that (i) the uniform Y_{Sa} assumption contributes at most ±3 pp to the standard-choice variance (§4.11), (ii) the R_z–R_a conversion uncertainty affects no reported variance fraction by more than 1 pp, and (iii) removing the ISO R_z cap changes the interaction term by +0.1 pp. These sensitivity analyses are intended to pre-empt reviewer concerns about modeling simplifications and to provide a template for analogous checks when the framework is applied to new gear geometries or materials.

References

[1] ISO 6336-3:2019. Calculation of load capacity of spur and helical gears—Part 3: Calculation of tooth bending strength. ISO, Geneva.

[2] AGMA 2001-D04. Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. American Gear Manufacturers Association, 2004.

[3] Forschungskuratorium Maschinenbau (FKM). Analytical Strength Assessment of Components, 7th ed. VDMA Verlag, 2020.

[4] Stephens R.I., Fatemi A., Stephens R.R., Fuchs H.O. Metal Fatigue in Engineering, 2nd ed. Wiley, 2001. (ISBN: 978-0-471-51059-8)

[5] Dowling N.E. Mechanical Behavior of Materials, 4th ed. Pearson, 2013. (ISBN: 978-0-13-139506-0)

[6] Dowling N.E., Calhoun C.A., Arcari A. Mean stress effects in stress-life fatigue and the Walker equation. Fatigue Fract. Eng. Mater. Struct. 32(3): 163–179, 2009. doi:10.1111/j.1460-2695.2008.01310.x

[7] Fatemi A., Yang L. Cumulative fatigue damage and life prediction theories: a survey. Int. J. Fatigue 20(1): 9–34, 1998. doi:10.1016/S0142-1123(97)00081-9

[8] University of Waterloo, Fatigue Data Exchange (FDE). SMDIdbase: AISI 4140 Steel, Quenched & Tempered, Fatigue Reports—Iterations #66 and #68. https://fde.uwaterloo.ca/Fde/Materials/SMDIdbase/Steel/

[9] Wang J.J., Pei W.C., Ji H.C., Long H.Y., Wang Z.T. Research on bending fatigue test based on 42CrMo gear (in Chinese). J. Mech. Strength 45(2): 474–480, 2023. doi:10.16579/j.issn.1001.9669.2023.02.030

[10] Glodez S., Sraml M., Kramberger J. A computational model for determination of service life of gears. Int. J. Fatigue 24(10): 1013–1020, 2002. doi:10.1016/S0142-1123(02)00024-5

[11] Pagliari L., Hong I., Concli F. An experimental analysis of the gear tooth bending strength predictions from ISO 6336 in low and high cycle fatigue. Forsch. Ingenieurwes. 89: 160, 2025. doi:10.1007/s10010-025-00928-6

[12] Bonaiti L., Monti M., Geitner M., Tobie T., Gorla C., Stahl K. On the usage of pulsator data within the load spectra assessment of gears. Int. J. Fatigue 181: 108145, 2024. doi:10.1016/j.ijfatigue.2024.108145

[13] Vietze D., Pellkofer J., Stahl K. Study on the Potential of New Load-Carrying Capacity Descriptions for the Service Life Calculations of Gears. Machines 12(5): 304, 2024. doi:10.3390/machines12050304